

GAS GIANT

Jupiter's surface is not solid. Each light or dark band and every big or small swirl or spot is a part of the planet's cloudy atmosphere.

JUPITER PROFILE

AVERAGE DISTANCE FROM THE SUN 483.6 million miles (778.3 million km)	ROTATION PERIOD 9.93 hours
CLOUD-TOP TEMPERATURE -160°F (-110°C)	ORBITAL PERIOD (LENGTH OF YEAR) 11.86 Earth years
DIAMETER 88,846 miles (142,984 km)	MASS (EARTH = 1) 318
VOLUME (EARTH = 1) 1,321	CLOUD-TOP GRAVITY (EARTH = 1) 2.53
NUMBER OF MOONS 79	SIZE COMPARISON
OBSERVATION Jupiter is bright and easy to spot. It has a maximum magnitude of -2.9. Even at its faintest, it is brighter than Sirius, the brightest star in the sky. Jupiter is best seen at opposition, which occurs once every 13 months.	

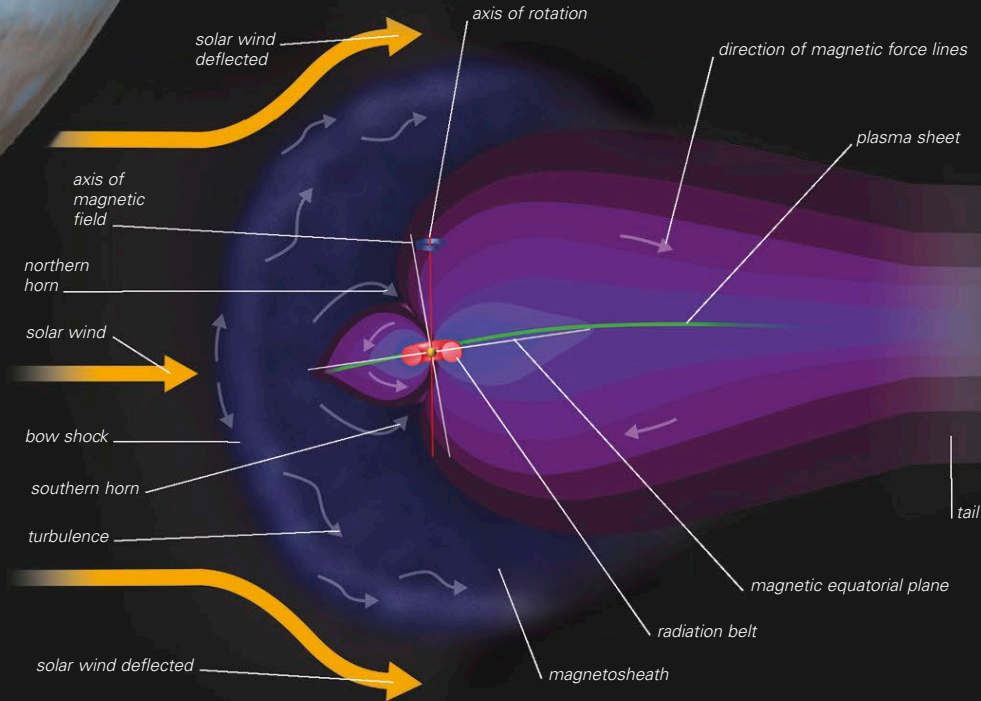
MAGNETIC FIELD

Jupiter has a magnetic field—it is as if the planet had a large bar magnet deep inside it. The field is generated by electric currents within the thick layer of metallic hydrogen, and the axis joining the magnetic poles is tilted at about 11° to the spin axis. The field is stronger than that of any other planet. It is about 20,000 times stronger than Earth's magnetic field and has great influence on the volume of space surrounding Jupiter. Solar wind particles (see p.107) streaming from the Sun plow into the field. They are slowed down and rerouted to spiral along the field's magnetic lines of force. Some particles enter Jupiter's upper atmosphere around its magnetic poles. They collide with the atmospheric gases, which radiate and produce aurorae. Other charged particles (plasma) are trapped and form a disklike sheet around Jupiter's magnetic equator. Electric currents flow within this sheet. High-energy particles are trapped and form radiation belts, similar to but much more intense than the Van Allen belts around Earth (see p.125). Jupiter's magnetic field is shaped by the solar wind, forming a vast region called the magnetosphere. Its size varies with changes in pressure of the solar wind, but the tail is thought to have a length of about 370 million miles (600 million km).



AURORA

This striking electric-blue aurora centered on Jupiter's north magnetic pole was imaged by the Hubble Space Telescope in 1998.



JUPITER'S MAGNETOSPHERE

Jupiter's magnetosphere—the bubblelike region around Jupiter dominated by the planet's magnetic field—is enormous; it is a thousand times the volume of the Sun, and its tail stretches away from the planet as far as Saturn's orbit. This slice through the magnetosphere reveals its structure.